ECE 340 # Homework Assignment #6 **Fall 2026**

DUE: FRIDAY, OCTOBER 16TH, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN ON TIME AT 5:00PM ON THE DUE DATE.

1. STEADY-STATE MINORITY-CARRIER DIFFUSION

Consider a long, uniformly doped n-type silicon sample at $T = 300$ K. The sample occupies $x \geq 0$ and may be treated as semi-infinite. The material parameters are

$$N_D = 1.0 \times 10^{16}\ \text{cm}^{-3}, \qquad n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}, \qquad \mu_p = 450\ \text{cm}^2/\text{V}\cdot\text{s},$$

and the minority-hole lifetime is $\tau_p = 2.0\ \mu$s. At $x = 0$, illumination maintains an excess hole concentration $\delta p(0) = 1.0 \times 10^{14}\ \text{cm}^{-3}$. For $x > 0$, there is no optical generation and no applied electric field. Assume steady-state conditions, low-level injection, and $\delta p(x) \to 0$ as $x \to \infty$. Use $kT/q = 0.0259$ V and $q = 1.60 \times 10^{-19}$ C.

(A). Calculate the equilibrium minority-hole concentration p_{n0}. Verify that the specified excess carrier concentration satisfies the low-level-injection condition.

(B). Use the Einstein relation to calculate the hole diffusion coefficient D_p and the hole diffusion length L_p.

(C). Starting with the steady-state minority-carrier diffusion equation

$$D_p \frac{d^2 \delta p}{dx^2} - \frac{\delta p}{\tau_p} = 0,$$

derive $\delta p(x)$. At what distance from $x = 0$ has the excess hole concentration fallen to 10% of its value at $x = 0$?

(D). Determine the hole diffusion current density $J_p(x)$. Calculate J_p at $x = 0$ and at $x = L_p$, and state the direction of the conventional hole current.

(E). Show that the total recombination current per unit area in the region $x > 0$,

$$J_{\text{rec}} = q \int_0^\infty \frac{\delta p(x)}{\tau_p}\, dx,$$

equals the hole diffusion current entering the region at $x = 0$. Briefly explain the physical meaning of this result.

Solution **20 points in total. If numerical answer, award half points if calculator error.**

(A). For nondegenerate n-type silicon with complete ionization, $n_0 \approx N_D$. From the mass-action law,

$$p_{n0} = \frac{n_i^2}{n_0} \approx \frac{n_i^2}{N_D} = \frac{(1.5\times10^{10})^2}{1.0\times10^{16}} = \boxed{2.25\times10^4\ \text{cm}^{-3}}.$$

Low-level injection in n-type material requires $\delta p \ll n_0 \approx N_D$. At the point of maximum injection,

$$\frac{\delta p(0)}{N_D} = \frac{1.0\times10^{14}}{1.0\times10^{16}} = 0.01.$$

The injected concentration is only 1% of the majority-electron concentration, so the low-level-injection approximation is valid.

4 points

(B). The Einstein relation gives

$$D_p = \mu_p \frac{kT}{q} = (450)(0.0259) = \boxed{11.7\ \text{cm}^2/\text{s}}.$$

Therefore,

$$L_p = \sqrt{D_p\tau_p} = \sqrt{(11.7)(2.0\times10^{-6})} = 4.83\times10^{-3}\ \text{cm} = \boxed{48.3\ \mu\text{m}}.$$

4 points

(C). Using $L_p^2 = D_p\tau_p$, the diffusion equation becomes

$$\frac{d^2\delta p}{dx^2} - \frac{\delta p}{L_p^2} = 0.$$

Its general solution is

$$\delta p(x) = A\exp(-x/L_p) + B\exp(x/L_p).$$

The boundary condition at infinity requires $B = 0$, while $\delta p(0) = 1.0\times10^{14}\ \text{cm}^{-3}$ gives $A = \delta p(0)$. Thus,

$$\boxed{\delta p(x) = 1.0\times10^{14}\exp(-x/L_p)\ \text{cm}^{-3}}.$$

For $\delta p(x) = 0.10\delta p(0)$,

$$0.10 = \exp(-x/L_p), \qquad x = L_p\ln(10) = (48.3\ \mu\text{m})(2.303) = \boxed{111\ \mu\text{m}}.$$

4 points

(D). Because $p(x) = p_{n0} + \delta p(x)$ and p_{n0} is constant,

$$J_p(x) = -qD_p \frac{dp}{dx} = -qD_p \frac{d\delta p}{dx} = \boxed{\frac{qD_p \delta p(0)}{L_p} \exp(-x/L_p)}.$$

At the illuminated boundary,

$$J_p(0) = \frac{(1.60\times10^{-19})(11.7)(1.0\times10^{14})}{4.83\times10^{-3}} = \boxed{3.87\times10^{-2}\ \text{A/cm}^2}.$$

At one diffusion length,

$$J_p(L_p) = J_p(0)e^{-1} = \boxed{1.42\times10^{-2}\ \text{A/cm}^2}.$$

Since $dp/dx < 0$, $J_p > 0$; the conventional hole current flows in the $+x$ direction.
4 points

(E). Substituting the excess-hole profile,

$$J_{\text{rec}} = \frac{q\delta p(0)}{\tau_p} \int_0^\infty \exp(-x/L_p)\, dx = \frac{q\delta p(0) L_p}{\tau_p}.$$

Because $L_p^2 = D_p \tau_p$,

$$\frac{L_p}{\tau_p} = \frac{D_p}{L_p},$$

and therefore

$$J_{\text{rec}} = \frac{qD_p \delta p(0)}{L_p} = J_p(0) = \boxed{3.87\times10^{-2}\ \text{A/cm}^2}.$$

In steady state, excess charge does not accumulate: every hole entering the $x > 0$ region by diffusion eventually recombines there. Thus the incoming diffusion current equals the integrated recombination current.
4 points

2. P-N JUNCTIONS IN EQUILIBRIUM

Consider a GaAs p-n diode that is asymmetrically doped with $N_D = 4 \times 10^{16} cm^{-3}$ and $N_A = 1.6 \times 10^{17} cm^{-3}$. Answer the following questions assuming the diode is in equilibrium at $T = 300K$. For GaAs, $E_g = 1.43eV$, $n_i = 2.1 \times 10^6 cm^{-3}$, and $\epsilon = 12.9\epsilon_0$.

(A). What is the equilibrium condition as it pertains to both the Fermi level and net current flow?

(B). Calculate the contact potential V_0.

(C). Calculate the depletion width in the n-region (x_{n0}), the depletion width in the p-region (x_{p0}), and the overall depletion width (W).

(D). Calculate the electrostatic potential **relative to the p-side** at the metallurgic junction $V(x = 0)$.

(E). Plot the electric field $E(x)$ as a function of x and calculate the peak value.

(F). Plot the band diagram paying attention to the shape of the bands in the junction and labeling E_i, E_C, E_V, and E_F.

Solution **20 points in total. If numerical answer, award half points if calculator error.**

(A). In equilibrium there is no net current flow and the Fermi level is flat.
3 points

(B).

$$V_{bi} = \frac{kT}{q} ln\left(\frac{N_D N_A}{n_i^2}\right) = 1.26V \tag{2.1}$$

3 points

(C).

$$W = \sqrt{\frac{2\epsilon V_{bi}}{q}\frac{N_A + N_D}{N_A N_D}} = 237nm \tag{2.2}$$

$$x_{n0} = W\frac{N_A}{N_A + N_D} = 189.6nm \tag{2.3}$$

$$x_{p0} = W - x_{n0} = 47.4nm \tag{2.4}$$

3 points

(D).

$$V(x=0) = -\int_{-x_{p0}}^{0} E(x)dx = -\frac{1}{2}E_0 x_{p0} = \frac{1}{2}\frac{q}{\epsilon}N_A x_{p0} x_{p0} = 0.252V \tag{2.5}$$

3 points

(E).

$$E_0 = -\frac{q}{\epsilon}N_A x_{p0} = -106kV/cm \tag{2.6}$$

See Fig. 2.

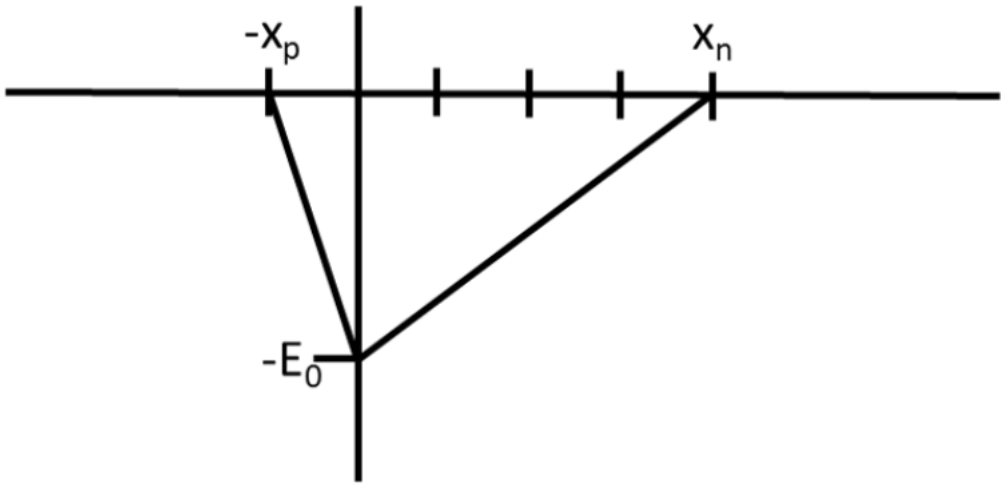


Figure 2.1: Electric field profile

4 points

(F). See Fig. 2

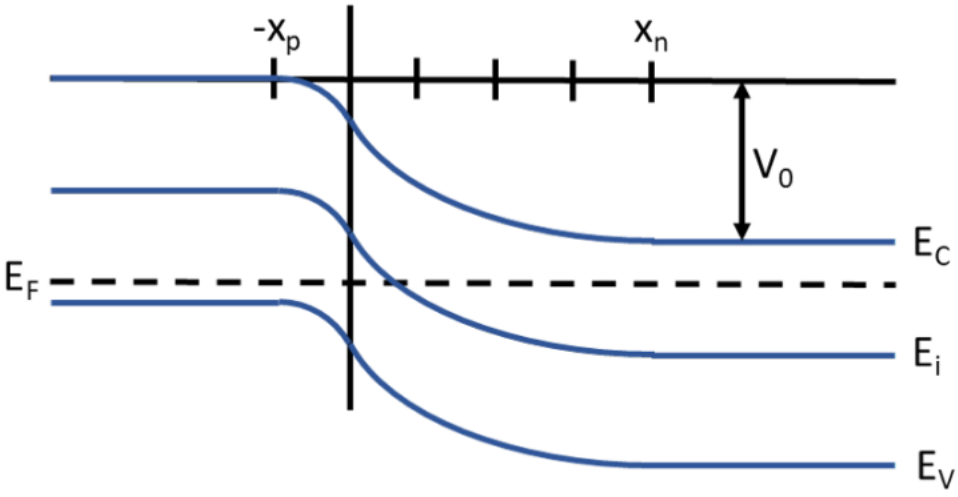


Figure 2.2: Bandstructure of the junction

4 points

3. JUNCTION PROPERTIES

Consider two p^+n junctions. One has an n-doping of N_D and the other has an n-doping of $\frac{1}{2}N_D$. Everything else remains unchanged. How do the following quantities change in the second junction compared to the first one? Indicate only increase or decrease and explain your answer for each **mathmatically**.

(A). Total amount of depletion charge.

(B). Magnitude of the peak electric field.

(C). Depletion width on the n-side (x_n).

(D). Depletion width on the p-side (x_p).

(E). Built-in potential (V_0).

Solution **20 points in total. Each question is worth total 4 points. 2 points for correct answer of increase or decrease. 2 points for valid explanation. At a minimum a valid explanation can just be a proportionality argument paired with valid steps arriving at the proportionality equation.**

(A). Decreases.

$$\begin{aligned} Q &= N_D x_n + N_A x_p \\ &= N_D \sqrt{\frac{2\epsilon V_{bi}}{q}\frac{N_A}{N_D(N_A+N_D)}} + N_A \sqrt{\frac{2\epsilon V_{bi}}{q}\frac{N_D}{N_A(N_A+N_D)}} \\ &= 2\sqrt{\frac{2\epsilon V_{bi}}{q}\frac{N_A N_D}{(N_A+N_D)}} \qquad (3.1) \\ (N_A \gg N_D) &\approx 2\sqrt{\frac{2\epsilon V_{bi}}{q} N_D} \qquad (3.2) \end{aligned}$$

If N_D is halved, V_{bi} decreases slightly (see part (E)), and since $Q \propto \sqrt{V_{bi} N_D}$, the total depletion charge decreases overall.

(B). Decreases.

$$\begin{aligned} |E_0| &= \frac{q}{\epsilon} N_D x_n \\ &= \sqrt{\frac{2qV_{bi}}{\epsilon}\frac{N_A N_D}{N_A+N_D}} \qquad (3.3) \\ (N_A \gg N_D) &\approx \sqrt{\frac{2qV_{bi}}{\epsilon} N_D} \qquad (3.4) \end{aligned}$$

If N_D is halved, $E_0 \propto \sqrt{V_{bi} N_D}$ decreases.

(C). Increases.

$$x_n = \sqrt{\frac{2\epsilon V_{bi}}{q}\frac{N_A}{N_D(N_A+N_D)}} \qquad (3.5)$$

$$(N_A \gg N_D) \approx \sqrt{\frac{2\epsilon V_{bi}}{qN_D}} \qquad (3.6)$$

Halving N_D increases x_n approximately by $\sqrt{2}$, because the depletion extends farther into the more lightly doped n-side.

(D). Decreases.

$$x_p = \sqrt{\frac{2\epsilon V_{bi}}{q}\frac{N_D}{N_A(N_A+N_D)}} \tag{3.7}$$

$$(N_A \gg N_D) \approx \sqrt{\frac{2\epsilon V_{bi}}{q}}\frac{\sqrt{N_D}}{N_A} \tag{3.8}$$

Since $x_p \propto \sqrt{N_D}$, halving N_D decreases x_p by approximately $\sqrt{1/2}$.

(E). Decreases.

$$V_{bi} = \frac{kT}{q}\ln\left(\frac{N_A N_D}{n_i^2}\right) \tag{3.9}$$

If N_D is halved, $V'_{bi} = V_{bi} - \frac{kT}{q}\ln 2$, so the built-in potential decreases slightly.

4. Si n–p Junction in Equilibrium

An *abrupt* silicon **n–p** (note the order) junction has a circular cross–section of diameter $10\,\mu$m. The doping and material parameters are:

$$N_D = 4.0\times10^{16}\ \text{cm}^{-3}, \qquad N_A = 1.5\times10^{17}\ \text{cm}^{-3}, \qquad n_i = 1.5\times10^{10}\ \text{cm}^{-3},$$

and the silicon permittivity is $\varepsilon_s = \varepsilon_{\text{Si}} = 11.8\,\varepsilon_0$. Assume $T = 300\,\text{K}$ unless otherwise specified.

(A). Calculate the Fermi–level positions in each quasi–neutral region at equilibrium: $(E_F - E_i)$ in the n region and $(E_i - E_F)$ in the p region. Then compute the contact (built-in) potential V_0 of the junction.

(B). Sketch the equilibrium energy-band diagram of the **n–p** junction (conduction band, valence band, intrinsic level, Fermi level, and built-in potential). Clearly label E_C, E_V, E_i, E_F, and qV_0.

(C). Determine the depletion width W, the depletion extents x_{n0} and x_{p0} on the n and p sides, respectively, the total uncovered ionized charge Q_+ in the depletion region, and the maximum electric field magnitude E_0. Use $\varepsilon_s = 11.8\,\varepsilon_0$ and take q as the elementary charge.

Solution **20 points total.**

(A). The intrinsic carrier concentration of Si at $T = 300\,K$ is $n_i = 1.5\times10^{10}\,\text{cm}^{-3}$ and charge neutrality $n_0 + N_A^- = p_0 + N_D^+$ must be satisfied.

(I) N–SIDE: Assuming all donors are thermalized, we can say $N_D^+ = 4\times10^{16}\,\text{cm}^{-3}$, and since this is an n–type material, $N_A^- = 0$. Also, since the thermalized donor level is far greater than n_i, we can safely assume that p_0 can be neglected, resulting in $n_0 = 4\times10^{16}\,\text{cm}^{-3}$.

Starting with:

$$n_0 = n_i \exp\left(\frac{E_{Fn} - E_i}{k_B T}\right) \tag{4.1}$$

Solving for $E_{Fn} - E_i$:

$$E_{Fn} - E_i = k_B T \ln\left(\frac{n_0}{n_i}\right) \tag{4.2}$$

$$E_{Fn} - E_i = 0.0259 \ln\left(\frac{4\times10^{16}}{1.5\times10^{10}}\right) = \boxed{0.383\,\text{eV}}$$

(II) P–SIDE: Assuming all acceptors are thermalized, we can say $N_A^- = 1.5\times10^{17}\,\text{cm}^{-3}$ and $N_D^+ = 0$. Also, since the thermalized acceptor level is far greater than n_i, we can safely assume that n_0 can be neglected, resulting in $p_0 = 1.5\times10^{17}\,\text{cm}^{-3}$.

Starting with:

$$p_0 = n_i \exp\left(\frac{E_i - E_{Fp}}{k_B T}\right) \tag{4.3}$$

Solving for $E_i - E_{Fp}$:

$$E_i - E_{Fp} = k_B T \ln\left(\frac{p_0}{n_i}\right) \tag{4.4}$$

$$E_i - E_{Fp} = 0.0259 \ln\left(\frac{1.5\times10^{17}}{1.5\times10^{10}}\right) = \boxed{0.417\,\text{eV}}$$

There are multiple ways to calculate the contact potential. For example:

$$qV_0 = (E_{Fn} - E_i) + (E_i - E_{Fp}) = 0.383 + 0.417 = \boxed{0.800\,\text{eV}}$$

or equivalently:

$$qV_0 = k_B T \ln\left(\frac{N_A N_D}{n_i^2}\right) = 0.0259 \ln\left(\frac{4\times10^{16}\times1.5\times10^{17}}{(1.5\times10^{10})^2}\right) = \boxed{0.800\,\text{eV}}$$

Solution **6 points total. Half credit for calculator error.**

(B). Please note that you were asked to draw the band diagram assuming an **n–p** junction. Though this is physically no different than a p–n junction, your band diagrams will look reversed compared to those in the textbook. The point is to think about what you are representing instead of simply copying figures.

The band diagrams for the individual n– and p–sides are shown below. When the junction is formed, the Fermi level is constant over the entire structure in thermal equilibrium, while the conduction, valence, and intrinsic energy levels all bend as a result of the different constant potentials in the n and p regions.

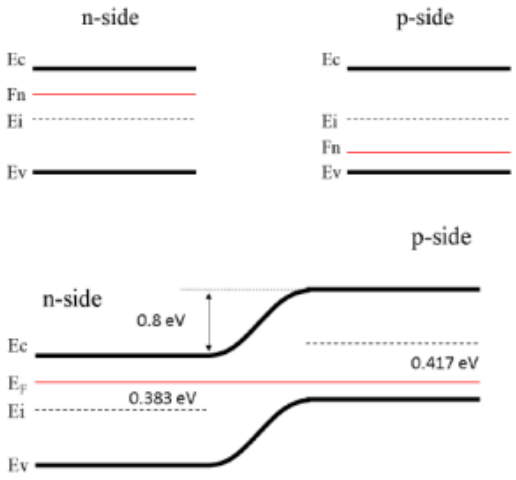


Figure 4.1: Band diagrams for n–p junction under equilibrium.

Solution **6 points total.**

(C). From Eqn. 5-21 of the textbook, the total depletion width can be calculated as:

$$W = \sqrt{\frac{2\varepsilon_s V_0}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)} \tag{4.5}$$

where $\varepsilon_s = \varepsilon_r \varepsilon_0 = (11.8)(8.85 \times 10^{-14})\,\text{F/cm}$.

$$W = \sqrt{\frac{2(11.8)(8.85 \times 10^{-14})(0.8)}{1.6 \times 10^{-19}}\left(\frac{1}{1.5 \times 10^{17}} + \frac{1}{4 \times 10^{16}}\right)} = \boxed{1.82 \times 10^{-5}\ \text{cm} = 0.182\,\mu\text{m}}$$

From Eqn. 5-23:

$$x_{n0} = W\frac{N_A}{N_A + N_D} = \boxed{0.144\,\mu\text{m}}, \quad x_{p0} = W\frac{N_D}{N_A + N_D} = \boxed{0.038\,\mu\text{m}}$$

Physically, this makes sense since x_{n0} must be larger than x_{p0} because the n–side has a lighter doping concentration.

From Eqn. 5-13:

$$Q_+ = qAx_{n0}N_D = (1.6\times10^{-19})\pi\left(\frac{10\times10^{-4}}{2}\right)^2(1.44\times10^{-5})(4\times10^{16}) = \boxed{7.22\times10^{-14}\,\text{C}}$$

From Eqn. 5-17, the maximum built–in electric field magnitude is:

$$E_0 = \frac{qx_{n0}N_D}{\varepsilon_s} = \frac{(1.6\times10^{-19})(1.44\times10^{-5})(4\times10^{16})}{(11.8)(8.85\times10^{-14})} = \boxed{8.80\times10^4\,\text{V/cm}}$$

$$\boxed{W = 0.182\,\mu\text{m}, \quad x_{n0} = 0.144\,\mu\text{m}, \quad x_{p0} = 0.038\,\mu\text{m}, \quad Q_+ = 7.22\times10^{-14}\,\text{C}, \quad E_0 = 8.80\times10^4\,\text{V/cm}}$$

Solution **10 points total. 2 points each quantity, half credit for calculator error.**